

Expt 5	Design of High Pass Filter using Windowing Techniques
Date:	5.1 Generate the coefficients of an FIR High-Pass filter using Rectangular windowing techniques.

Code:

```

clc;
clear;
close all;

%% =====
% FIR High-Pass Filter Design Using
% Rectangular, Hamming, and Hann Windows
%% =====

% Input Parameters
fp = 1000;      % High-pass cutoff frequency (Hz)
Fs = 8000;      % Sampling frequency (Hz)
N = 50;         % Filter order

% Normalized cutoff frequency
Wn = fp/(Fs/2);

%% 1. FIR High-Pass Filter using Rectangular Window
b_rect = fir1(N, Wn, 'high', rectwin(N+1));

disp('=====');
disp('High-Pass FIR Filter (Rectangular Window)');
disp('=====');
disp(b_rect);

%% 2. FIR High-Pass Filter using Hamming Window
b_hamm = fir1(N, Wn, 'high', hamming(N+1));

disp('=====');
disp('High-Pass FIR Filter (Hamming Window)');
disp('=====');
disp(b_hamm);

%% 3. FIR High-Pass Filter using Hann Window
b_hann = fir1(N, Wn, 'high', hann(N+1)); % Use hanning(N+1) if required

disp('=====');
disp('High-Pass FIR Filter (Hann Window)');
disp('=====');
disp(b_hann);

```

```

%%% Display Filter Information
fprintf('\nFilter Order      : %d\n', N);
fprintf('Sampling Frequency : %d Hz\n', Fs);
fprintf('Cutoff Frequency   : %d Hz\n', fp);
fprintf('Normalized Cutoff   : %.2f\n\n', Wn);

%%% Frequency Response - Rectangular Window
figure;
freqz(b_rect,1,1024,Fs);
title('High-Pass FIR using Rectangular Window');

%%% Frequency Response - Hamming Window
figure;
freqz(b_hamm,1,1024,Fs);
title('High-Pass FIR using Hamming Window');

%%% Frequency Response - Hann Window
figure;
freqz(b_hann,1,1024,Fs);
title('High-Pass FIR using Hann Window');

%%% Magnitude Response Comparison
[H1,f] = freqz(b_rect,1,1024,Fs);
[H2,~] = freqz(b_hamm,1,1024,Fs);
[H3,~] = freqz(b_hann,1,1024,Fs);

figure;
plot(f,abs(H1),'LineWidth',1.5);
hold on;
plot(f,abs(H2),'LineWidth',1.5);
plot(f,abs(H3),'LineWidth',1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude');
title('Comparison of FIR High-Pass Filters');
legend('Rectangular','Hamming','Hann');
xlim([0 Fs/2]);

```

Output:

Fig 1:

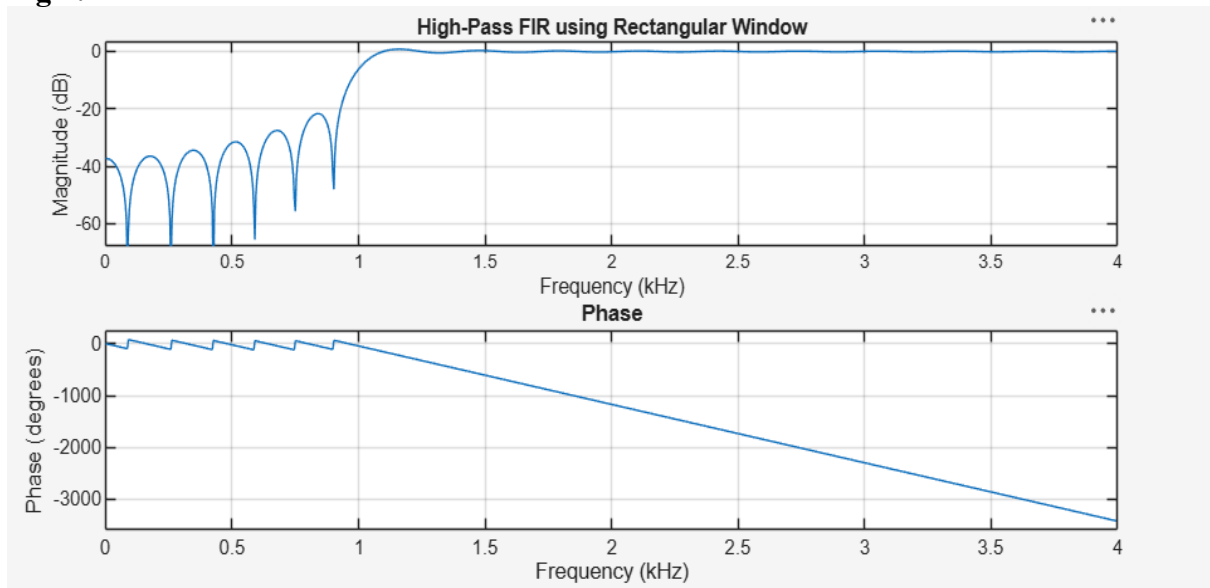


Fig 2:

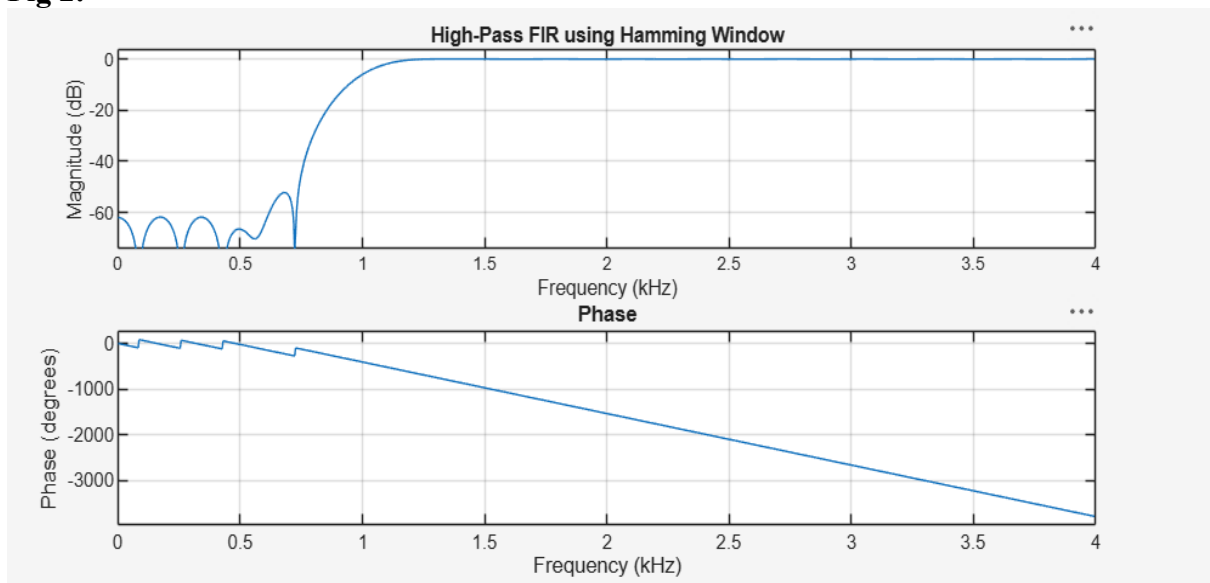


Fig 3:

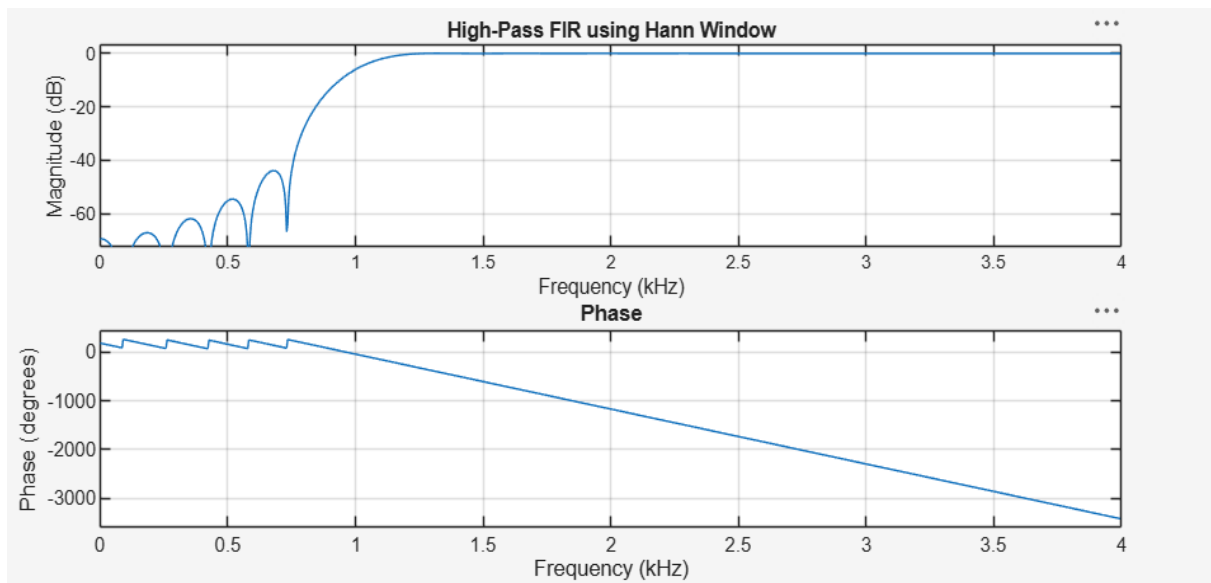
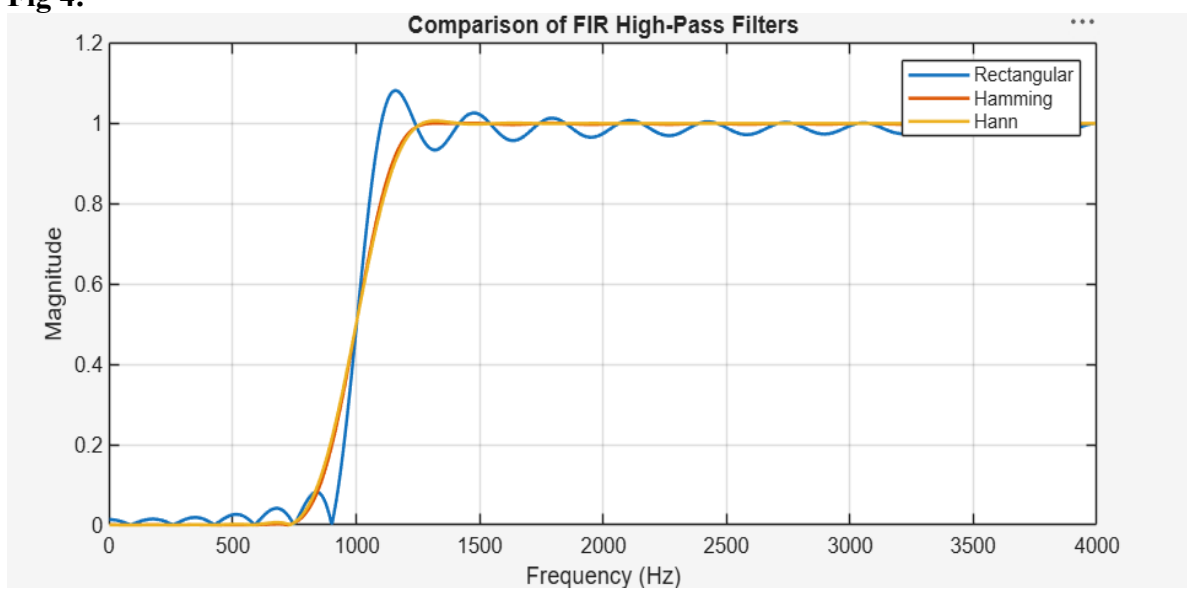


Fig 4:



Expt 5	Design of High Pass Filter using Windowing Techniques
Date:	5.2. Generate the coefficients of an FIR High-Pass filter using Hamming windowing techniques

Code:

```

clc;
clear;
close all;

%% --- Input Parameters ---
fp = 1000;    % High-pass cutoff frequency (Hz)
fs = 8000;    % Sampling frequency (Hz)
N = 50;       % Filter order
wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- FIR High-Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));

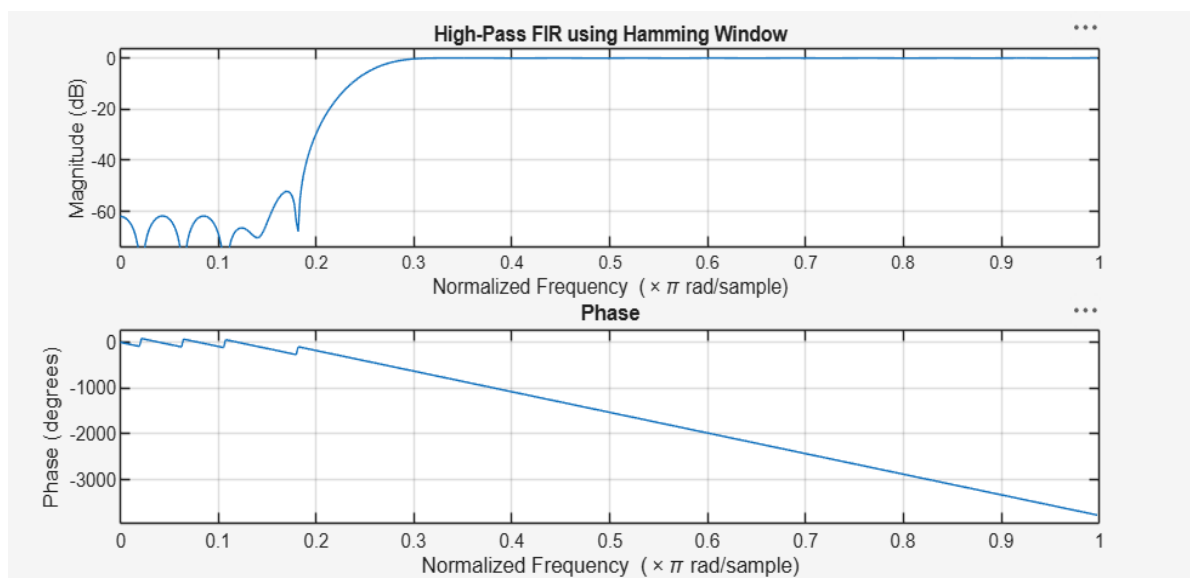
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm);

%% --- Frequency Response Plot ---
figure;
freqz(b_hamm,1);
title('High-Pass FIR using Hamming Window');

```

Output:

Fig 1:



Expt 5	Design of High Pass Filter using Windowing Techniques
Date:	5.3. Generate the coefficients of an FIR High-Pass filter using Hanning windowing techniques

Code:

```

clc;

clear;

close all;
%% --- Input Parameters ---
fp = 1000;    % High-pass cutoff frequency (Hz)
fs = 8000;    % Sampling frequency (Hz)
N = 50;       % Filter order
% Normalized cutoff frequency
wc = fp/(fs/2);
%% --- FIR High-Pass Filter using Hanning Window ---
b_hann = fir1(N, wc, 'high', hanning(N+1));
% Display filter coefficients
disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann);

%% --- Frequency Response ---
figure;
freqz(b_hann,1);
title('High-Pass FIR using Hanning Window');

```

Output:

Fig 1:

